

# TIC 309656476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

## BLS EPHEMERIS

Period 28.920445 d  
Epoch (t0) 2997.6132 BTJD  
Duration 0.1700 d  
Depth 0.050 % (495 ppm)  
SNR 24.6

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

## INDEPENDENT CHECKS (f3i gates)

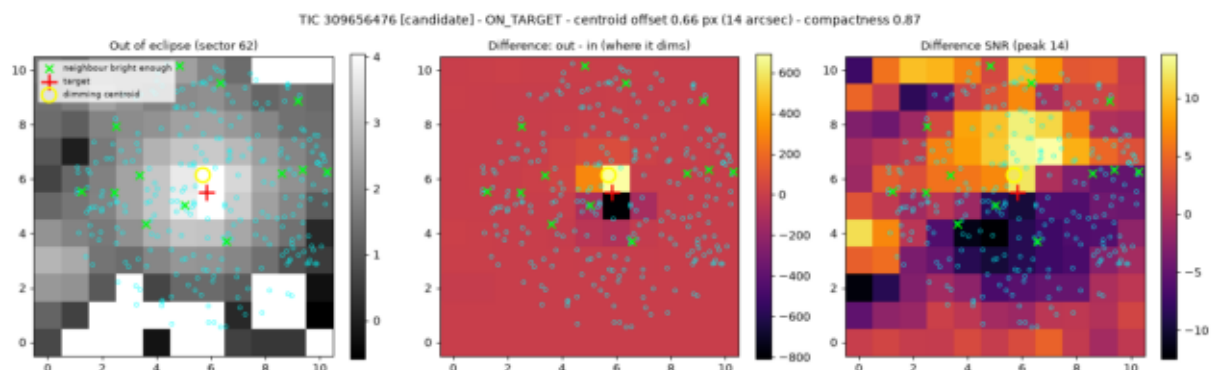
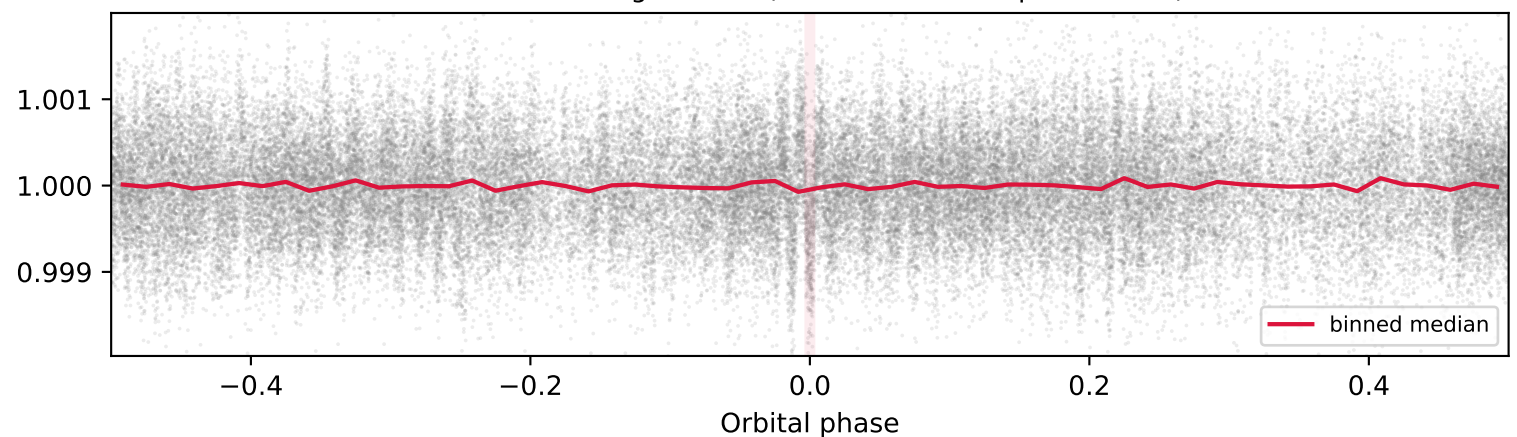
Eclipses observed 6 (gate  $\geq 3$ )  
Robust depth ratio 0.98 (gate  $\geq 0.50$ )  
Robust depth sigma 14.1 (gate  $\geq 7.0$ )  
Odd/even z-score 1.97 (consistent)  
Secondary eclipse not detected

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 62 light-curve dip in this sector: 8.5 sigma t0 refined by 0.3975 phase  
Difference-image peak SNR 13.80 (gate  $\geq 5.0$ ) Compactness 0.87 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.66 px (14 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 309656481  
1.29 px away, dTmag +5.82 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.